ITMO UNIVERSITY

LECTURE 2 · ITMO UNIVERSITY · 30 SEPTEMBER 2026

Social Media User Profiling

From text classifiers to multi-source profiles — and where profiling goes too far.

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Lecturer · Tutors



RECAP · YOUR HOMEWORK

What does the platform think it knows about you?



01

TOLD

What you typed in

Birthday, city, language, the accounts you follow, your bio and job title.

02

OBSERVED

What you did

Watched, clicked, liked, skipped, searched — and when, where and on which device.

03

INFERRED

What the model decided

Age band, interests, life events, “people like you who bought…”. This layer is profiling.

WHY THE THIRD LAYER MATTERS

The ad page shows you a handful of interest labels. The models behind them use hundreds of signals you never typed in — and by the end of today you will know how they are built and how accurate they get.

SOURCE Meta, “Why am I seeing this ad?”; Google My Ad Center; student homework from Lecture 1.

LECTURE 2 · OUTLINE

Five questions for today.

01

Who is the user?

What is a profile, and who wants one?

02

Text classification

Text in, label out: the building block.

03

Multi-source fusion

No platform sees the whole person.

04

Profiling in practice

From NUS-MSS to LLMs that profile in one prompt.

05

Where it goes too far

Pitfalls, ethics and the 2026 rules.

Lab 1 is released after today. Sections 2 and 3 are the toolkit you need for it.

ITMO UNIVERSITY

01

SECTION 1

Who is the user?

ETH ZURICH · ICLR 2024

GPT-4 read the Reddit comments of 520 people and guessed their personal details — right 85% of the time.

Top-3 accuracy: 95%. About 100× cheaper and 240× faster than human investigators. Profiling used to need a research team. Now it needs a prompt.



WHO IS THE USER · THE PROFILE

A user profile has four layers.



DEMOGRAPHIC

Who you are

Age, gender, occupation, education, income, family status. What advertisers buy.

PSYCHOGRAPHIC

How you think

Personality (the Big Five), values, interests, opinions.

BEHAVIOURAL & MOBILITY

What you do

Routines, places visited, content consumed, purchase intent.

AUTHENTICITY

Are you human?

Person, bot or AI agent? After Moltbook, this question comes first.

- ✓ **Stated.** Bio, birthday, job title — often missing or false.
- ✓ **Inferred.** Predicted from content and behaviour.
- ✓ **Scarce ground truth.** Most labels are self-reports or proxies.

SOURCE Farseev, ITMO SMA Module 9 (demography & mobility); Lecture 1 (Moltbook case).

WHO IS THE USER · APPLICATIONS

Six reasons to profile — and three ways to build one.

ADVERTISING

The car ad for likely car buyers; lookalike audiences.

RECOMMENDATION

The next video, product or friend to suggest.

URBAN PLANNING

Kindergartens where young families live; food courts near offices.

WELLNESS

Activity and lifestyle suggestions matched to routines.

SAFETY

Is this user a child? Age assurance is now a legal duty.

PUBLIC SECTOR & RESEARCH

Understand a population without running a survey.

THREE WAYS TO BUILD A PROFILE

01 **Classic: a stack of classifiers.** One model per attribute — age, gender, interests, bot or human — each trained and wrong on its own.

02 **Neural: one network, many heads.** Multi-task learning shares one user embedding and predicts all labels at once.

03 **LLM: out of the box.** Zero- or few-shot, every attribute from one prompt — for millions of users.

SOURCE Farseev, Nie, Akbari, Chua, “Harvesting Multiple Sources for User Profile Learning”, ACM ICMR 2015 (applications).

Age prediction: from ad feature to legal duty.



APR 2025	Meta starts using AI in the US to find teens who listed an adult birthday; moves them to Teen Accounts.
SEP 2025	Instagram's age AI expands to Canada, the UK and Australia.
10 DEC 2025	Australia's under-16 social media ban takes effect. Fines up to A$49.5M.
20 JAN 2026	OpenAI switches on age prediction in ChatGPT: activity times, usage patterns, stated age.
MAY 2026	Meta brings age AI to the EU, Brazil — and to Facebook. Flagged users verify by ID or face scan.

AUSTRALIA, FIRST MONTH

4.7M

under-16 accounts removed or restricted by 15 Jan 2026.

Lesson

Every false positive is an adult asked for ID. Every false negative is a child let in. Metric choice is policy.

SOURCE Meta Newsroom (21 Apr 2025; 5 May 2026); eSafety Commissioner release (15 Jan 2026); OpenAI, “Our approach to age prediction” (Jan 2026).

02

SECTION 2

Text classification: the basics



THE BUILDING BLOCK

Age, gender, personality, bot or human: every profile attribute is a label predicted from content.

Two old lectures in one section: feature types, social-text pre-processing, weighting, classic classifiers, embeddings, LLMs — and how to measure the result.



TEXT CLASSIFICATION · FEATURES

What counts as a text feature? Eight families.

01

Characters

Character n-grams survive typos, slang and hashtags: “noooo”, “#itmo”.

02

Words

Single terms — the classic bag of words. Simple and surprisingly strong.

03

N-grams & phrases

“not good”, “new job”: word order in small doses.

04

Named entities

People, places, organisations, brands, extracted by IE systems.

05

Linguistic units

Part-of-speech tags, dependency relations, sentence structure.

06

Lexicons

LIWC and sentiment dictionaries: counts of psychologically meaningful words.

07

Style & behaviour

Length, punctuation, emoji, URLs, mentions, misspellings, readability scores.

08

Embeddings

Dense vectors from a language model: meaning, not spelling. The 2026 default.

The old rule still holds: single-word features perform surprisingly well. Start there, then add families and measure what each one buys you.

SOURCE Farseev, ITMO SMA Module 2 (commonly used text features); Rangel et al., Overview of the Author Profiling Task at PAN 2014.

TEXT CLASSIFICATION · SOCIAL TEXT

A post is not a document: short, informal, live.



Short vs long	A post is a sentence fragment; a document has context built in.
Informal vs formal	Typos, abbreviations, phonetic spellings, emoji, invented words: “kewl”.
Conversational vs standalone	Replies and threads: one post often needs its neighbours to make sense.
Stream vs collection	Data arrives continuously, users come and go, vocabulary drifts.

PRE-PROCESSING FOR POSTS

- ✓ **Identify the language** before anything else.
- ✓ **Normalise:** “se u 2morw!!!” → see you tomorrow; b4 → before.
- ✓ **Filter noise:** URLs, @mentions, RT prefixes, stray symbols.
- ✓ **Drop empties:** posts under 4 characters after cleaning.

THEN AND NOW

2012: a normalisation dictionary of about 40,000 slang-to-standard pairs, mined from 80 million tweets. 2026: an LLM normalises in context — check it doesn't “fix” the meaning away.

SOURCE Farseev, ITMO SMA Module 3 (microblog classification); Han, Cook, Baldwin, “Automatically constructing a normalisation dictionary for microblogs”, EMNLP-CoNLL 2012.

TEXT CLASSIFICATION · THE PIPELINE

A computer classifies numbers, not words. Step one: text to vector.

01

Tokenise

Split into words: started · my · new · job · at · itmo · today

→

02

Stop list

Drop low-meaning words: my, at, the, a, of.

→

03

Stem / lemmatise

started → start, jobs → job. Fewer, denser features.

→

04

Weight

How much does each word matter to this document? (tf.idf, next slide)

→

05

Vector

One number per vocabulary word: [0, 0.8, 0, 1.2, ...]

THE SAME POST, END TO END

"Started my new job at ITMO today!!" → start · new · job · itmo · today → a sparse vector over the whole vocabulary, ready for any classifier.

WHAT CHANGED BY 2026

Subword tokenisers and dense embeddings replace the hand steps. Text in, vector out, classifier on top — that idea has not changed.

SOURCE Farseev, ITMO SMA Module 2 — Text Analysis and Classification (free text, stop lists, stemming).

PIPELINE · STEP 1: TOKENISE

Where does a word end? It depends on the tokeniser.

POST **“Can't wait to start @ITMO!!! 🚀 #NewJob”**

WHITESPACE

Can't	wait	to	start	@ITMO!!!	🚀	#NewJob

SOCIAL-AWARE RULES

can't	wait	to	start	@itmo	!!!	🚀	#newjob

SUBWORD (BPE)

Can	't	wait	to	start	@	IT	MO	!!!	🚀	#	New	Job

Illustrative splits; each tool differs in detail.

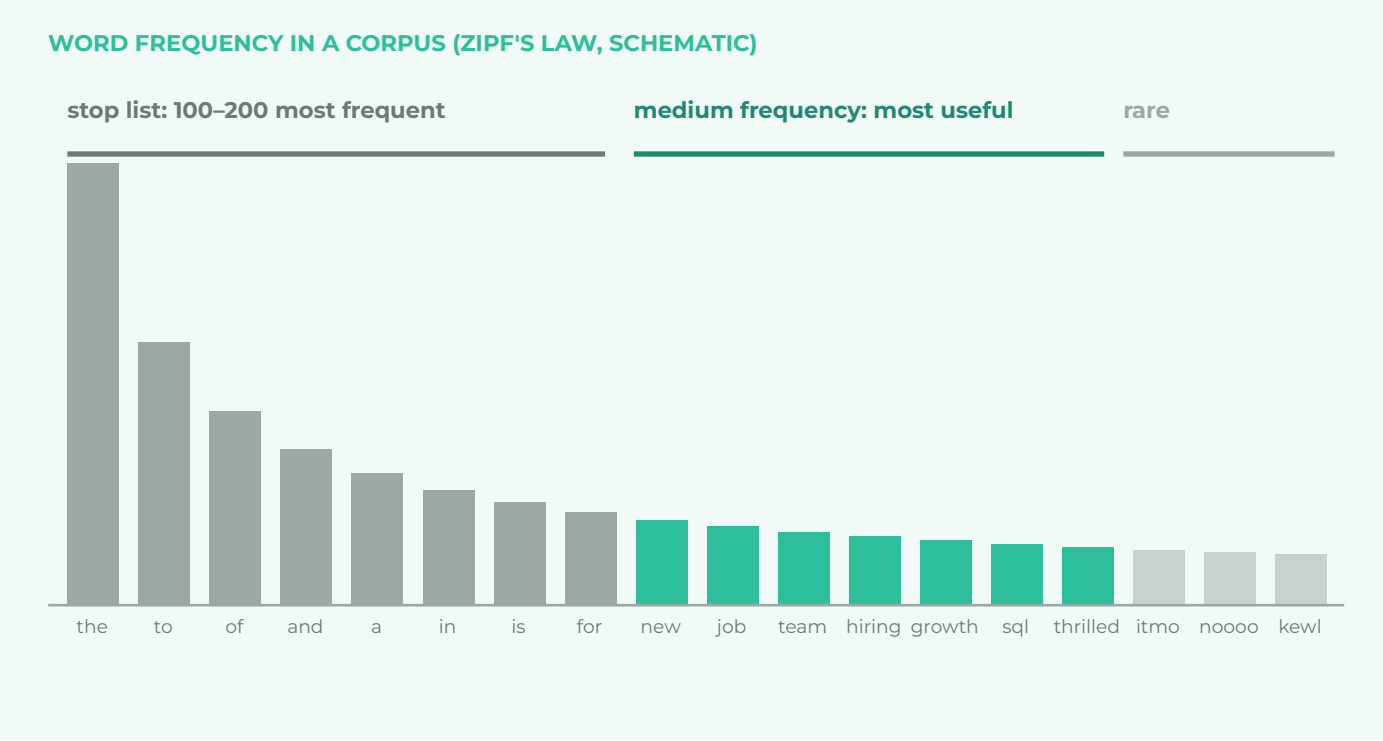
WHY IT MATTERS

- ✓ **Social text:** keep @mentions, #hashtags, emoji and “!!!” — they carry signal.
- ✓ **No spaces:** Chinese and Japanese need word segmentation first.
- ✓ **Subwords:** no word is ever unknown — “ITMO” becomes known pieces.
- ✓ **LLMs:** tokens are the unit of cost and of context length.

SOURCE Manning, Raghavan & Schütze, Introduction to Information Retrieval (Stanford / CUP 2008), Ch. 2; Sennrich, Haddow, Birch, “Neural Machine Translation of Rare Words with Subword Units”, ACL 2016; Farseev, ITMO SMA Module 3.

PIPELINE · STEP 2: STOP LIST

Drop the words that tell you nothing — carefully.



RULES OF THUMB

- ✓ **High-frequency words** don't discriminate; low-to-medium ones raise precision.
- ✓ **No universal list:** NLTK and Snowball ship lists per language.
- ✓ **Keep negations** for sentiment: “not good” ≠ “good”.
- ✓ **Phrase search** and pronoun-based features (LIWC's “I” vs “we”) need the stop words.
- ✓ **Embeddings and LLMs** use every token — skip this step for them.

SOURCE Manning, Raghavan & Schütze, IIR Ch. 2.2.2; Zipf (1949); Farseev, ITMO SMA Module 2 (term selection) and Module 3 (stop-word resources).

PIPELINE · STEP 3: STEM OR LEMMATISE

Many surface forms, one term — with side effects.

WORD	PORTER STEM	LEMMA	
recognize / recognized	**recogn**	recognize	
recognise	**recognis**	recognise	spelling not merged
studies	**studi**	study	
running	**run**	run	
better	**better**	good (adj.)	needs POS
organization	**organ**	organization	over-stemmed
university / universe	**univers**	two lemmas	meanings merged

Social text: slang breaks both (“goooood”, “b4”) — normalise with a dictionary first (Module 3).

STEMMING

Fast suffix rules

Porter (1980) strips endings like -ing, -ation, -ies. No dictionary, so it over- and under-stems — and can merge meanings.

LEMMATISATION

Dictionary + grammar

Returns the real base form, using the part of speech. Slower, exact — and essential for Russian, where a noun has up to 12 case-and-number forms.

SOURCE Porter, “An algorithm for suffix stripping”, Program 14(3), 1980 — stems computed with NLTK's PorterStemmer; IIR Ch. 2.2.4; Farseev, ITMO SMA Modules 2–3.

PIPELINE · STEPS 4–5: WEIGHT AND VECTORISE

A worked example: three posts, one query.

Q “information retrieval”

D1 “information retrieved by VS retrieval methods”

D2 “information theory forms the basis for probabilistic methods”

D3 “he retrieved his money from the safe”

	info	retriev	method	theory	VS	form	basis	probabil	money	safe
Q	1	1	0	0	0	0	0	0	0	0
D1	1	2	1	0	1	0	0	0	0	0
D2	1	0	1	1	0	1	1	1	0	0
D3	0	1	0	0	0	0	0	0	1	1

$SIM(Q, D) = \Sigma Q_i \cdot D_i$

D1 3

D2 1

D3 1

D1 wins. But D3 — about money — ties with D2 only because the stemmer turned “retrieved” into “retriev”. Weighting (tf.idf, next) and embeddings fix much of this.

SOURCE Worked example from Farseev, ITMO SMA Module 2 (“A retrieval example”); IIR Ch. 6 (vector space model).

TEXT CLASSIFICATION · WEIGHTING

tf.idf: frequent here, rare elsewhere, means important.

THE WEIGHT OF TERM T IN DOCUMENT D

$$w(t, d) = tf(t, d) \times (\log_2(N / df(t)) + 1)$$

tf — how often t appears in this document.
idf — log of (all documents ÷ documents containing t), plus one. Common words sink towards 1.

WORKED EXAMPLE: N = 1,000 DOCUMENTS

Term	df(t)	idf = $\log_2(N/df) + 1$
ALPHA	100	**4.32**
BETA	500	**2.00**
GAMMA	900	**1.15**

Rare terms get the highest weights.

VECTOR SPACE MODEL · COSINE SIMILARITY

$\cos(d_1, d_2) = (d_1 \cdot d_2) / (\|d_1\| \|d_2\|)$. Similarity is the angle between vectors, so two posts on the same topic match even if one is ten times longer. Salton's idea from the 1970s–80s — still the baseline to beat.

SOURCE Salton & Buckley, “Term-weighting approaches in automatic text retrieval”, IP&M 1988; Farseev, ITMO SMA Module 2 (idf example).

TEXT CLASSIFICATION · FEATURE SELECTION

Fewer, better features beat more, noisier ones.

- ✓ **Document frequency.** Drop words that almost never appear.
- ✓ **Chi-square (χ^2).** Keep words whose presence depends on the class.
- ✓ **Information gain.** Keep words that most reduce uncertainty about the label.
- ✓ **Projection (PCA).** Compress many correlated features into a few dimensions.
- ✓ **The curse of dimensionality.** More features need more data — or the model memorises noise.

YANG & PEDERSEN, ICML 1997

16,039 → 321

words kept after information-gain selection — 2% of the vocabulary.

87.9% → 89.2%

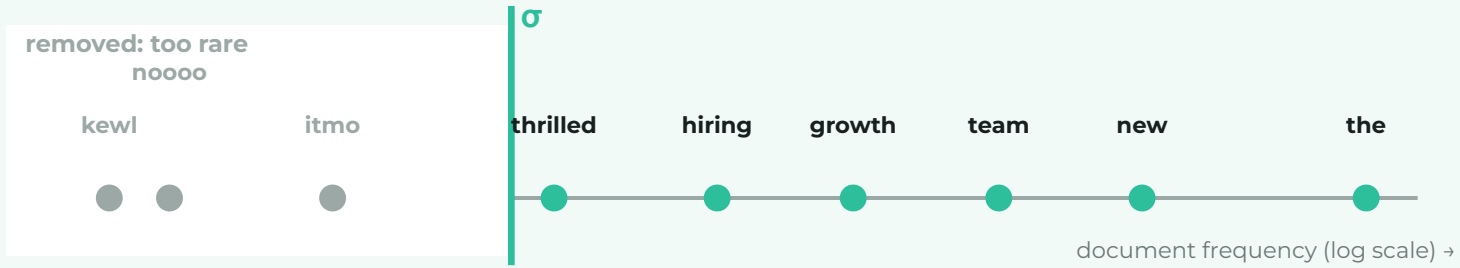
precision went up, not down. Lab 1's JSON has hundreds of fields; most won't help.

SOURCE Yang & Pedersen, “A comparative study on feature selection in text categorization”, ICML 1997; Farseev, ITMO SMA Module 2.

FEATURE SELECTION · DOCUMENT FREQUENCY

Keep a word only if enough posts use it.

KEEP TERM T IF DF(T) > Σ



RETRIEVAL (IR)

favours rare terms: high idf finds the one right document.

CLASSIFICATION (TC)

needs terms common within a class: a word seen twice can't define one.

VERDICT

Simple, scalable — and as good as χ^2 and IG.

- ✓ **Up to 90%** of terms removed with no loss, or a gain.
- ✓ **Ad hoc:** σ is set by trial on training data.
- ✓ **In code:** min_df in scikit-learn's vectorisers.

SOURCE Yang & Pedersen, ICML 1997; Farseev, ITMO SMA Module 2 (document frequency thresholding). Term frequencies are illustrative.

FEATURE SELECTION · χ^2 STATISTIC

χ^2: does the word depend on the class?

CONTINGENCY TABLE FOR TERM T, CLASS C

	in class c	not in c
t present	A	B
t absent	C	D

$$\chi^2(t, c) = N \cdot (AD - CB)^2 / [(A+C)(B+D)(A+B)(C+D)]$$

0 when t and c are independent; larger means more dependent.

WORKED EXAMPLE: 1,000 LINKEDIN POSTS, 100 ANNOUNCE A NEW JOB

Term	A	B	C	D	χ^2
"thrilled"	40	10	60	890	**286.5**
"today"	20	180	80	720	**0.0**

"today" appears in 10% of job posts and 10% of the rest — independent, $\chi^2 = 0$. "thrilled" is 8× more likely in a job post.

- ✓ **Rank terms per class** and keep the top k.
- ✓ **Comparable** only across terms for the same class.
- ✓ **Unreliable** for rare terms — combine with a df threshold.

SOURCE Yang & Pedersen, ICML 1997 (χ^2 term selection); IIR Ch. 13.5.2; Farseev, ITMO SMA Module 2. Counts are illustrative; χ^2 computed exactly.

FEATURE SELECTION · INFORMATION GAIN

Information gain: how much uncertainty does a split remove?

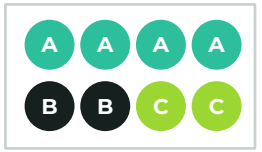
ENTROPY OF THE LABELS

$$H = -\Sigma\, p \cdot \log_2 p$$

INFORMATION GAIN OF A SPLIT

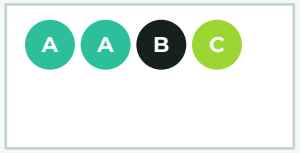
IG = H(before) − weighted H(after)

Zero if the split leaves the mix unchanged. Large if it produces purer groups. The same quantity picks the splits in decision trees.



8 examples: H = 1.5 bits

SPLIT ON "TYPE"



H = 1.5

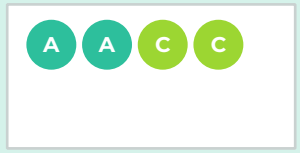


H = 1.5

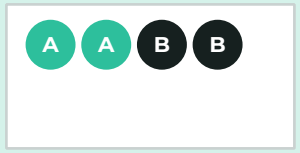
IG = 0 bits

useless split

SPLIT ON "PATRONS"



H = 1.0



H = 1.0

IG = 0.5 bits

keep this split

SOURCE Worked example from Farseev, ITMO SMA Module 2 (after Russell & Norvig, AI: A Modern Approach); Yang & Pedersen, ICML 1997.

FEATURE SELECTION · PROJECTION AND A VERDICT

Select words, or compress them? Pick by what you need to explain.

PCA / SVD: ROTATE, THEN KEEP THE TOP AXES



New features are mixtures of words: compact, but no longer readable. On text, truncated SVD is known as LSA.

METHOD	USES LABELS	READABLE	VERDICT (YANG & PEDERSEN, 1997)
Doc. frequency	no	yes	As good as IG and χ^2; drops up to 90% of terms.
χ^2	yes	yes	Strong; unreliable on rare terms.
Information gain	yes	yes	Strong: 16,039 → 321 terms, precision up.
Mutual information	yes	yes	Weak: favours rare terms.
Term strength	no	yes	Global, not per class; mid-pack.
PCA / SVD	no	no	Compact, but loses readable features.

SOURCE Yang & Pedersen, “A comparative study on feature selection in text categorization”, ICML 1997; Deerwester et al., JASIS 1990 (LSA); Farseev, ITMO SMA Module 2.

TEXT CLASSIFICATION · BEYOND WORDS

Text alone is not enough: four more feature families.

01

Evolving text

Vocabulary drifts. Weight recent terms higher and retrain on a sliding window; 24 hours worked best.

02

Location

“ITMO” from St Petersburg is the university; from Italy, a medical-trials group. Use profile, time zone, geo-tags.

03

Social relations

Follows, replies, retweets. 80% of relevant users sat within two hops of a known account.

04

Posting tendency

70–80% of tweets on a topic never name it. The author's last posts reveal what this one is about.

RESULT

Text features alone were not effective. Each added family raised F1, and all four together were best.

SOURCE Chen, Li, Chua, “Real-time Filtering of Event-Related Microblogs”, NUS report 2014; Farseev, ITMO SMA Module 3.

TEXT CLASSIFICATION · CLASSIC MODELS

Five classic classifiers. Know what each is good for.

01

Naive Bayes

Word counts per class, multiplied as probabilities.

GOOD FOR

Short text, little data, fast baselines.

02

k-nearest neighbours

Find the k most similar labelled documents; let them vote.

GOOD FOR

Small datasets; explainable by example.

03

SVM

The boundary with the widest margin between classes.

GOOD FOR

High-dimensional sparse text. The 2000s default.

04

Decision tree

A chain of yes / no questions on features.

GOOD FOR

Readable rules; mixed feature types.

05

Random forest

Many trees on random slices of data, voting.

GOOD FOR

Robust, little tuning. Wins the profiling study later.

✓ **Ensembles:** bagging, boosting and stacking combine weak models into a strong one — the same idea as late fusion in section 3.

SOURCE Yang & Liu, “A re-examination of text categorization methods”, SIGIR 1999; Farseev, ITMO SMA Module 2.

Naive Bayes: flip the question with Bayes' rule.

BAYES' RULE

$$p(c \mid d) = p(d \mid c) \cdot p(c) / p(d)$$

p(boy | named Jon)	**1**
p(named Jon)	**0.01**
p(boy)	**0.5**

p(named Jon | boy) = 0.01 × 1 / 0.5 = 0.02

Hard to estimate directly, easy to compute from counts.

FOR TEXT

$$\hat{c} = \text{argmax}\ p(c) \cdot \prod p(w_i \mid c)$$

- ✓ **Naive:** words independent given the class.
- ✓ **Drop p(d):** it is the same for every class.
- ✓ **Smooth:** add-one, so an unseen word can't zero the product.
- ✓ **Use logs:** sum log-probabilities, avoid underflow.

WHY IT STILL MATTERS

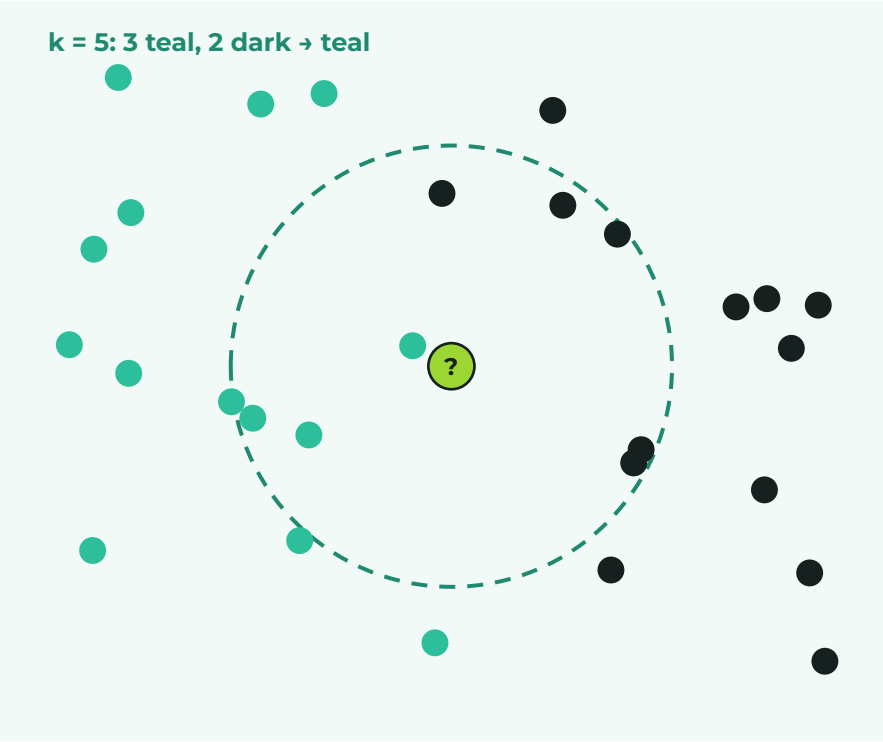
Seconds to train. Every word's vote is readable.

Needs little data and is hard to overfit badly — a fair first baseline for Lab 1, before any embedding or LLM.

SOURCE Farseev, ITMO SMA Module 2 (Bayes example); Jurafsky & Martin, Speech and Language Processing (3rd ed. draft, Aug 2026), Appendix B.

CLASSIFIERS · K-NEAREST NEIGHBOURS

kNN: you are the company your posts keep.



01 Measure

cosine similarity between the new post and every training post.

02 Pick

the k most similar (Yang & Liu used k = 45).

03 Vote

score each class by summing its neighbours' similarities.

04 Threshold

tune a per-class cut-off on an 80 / 20 split, maximising F1.

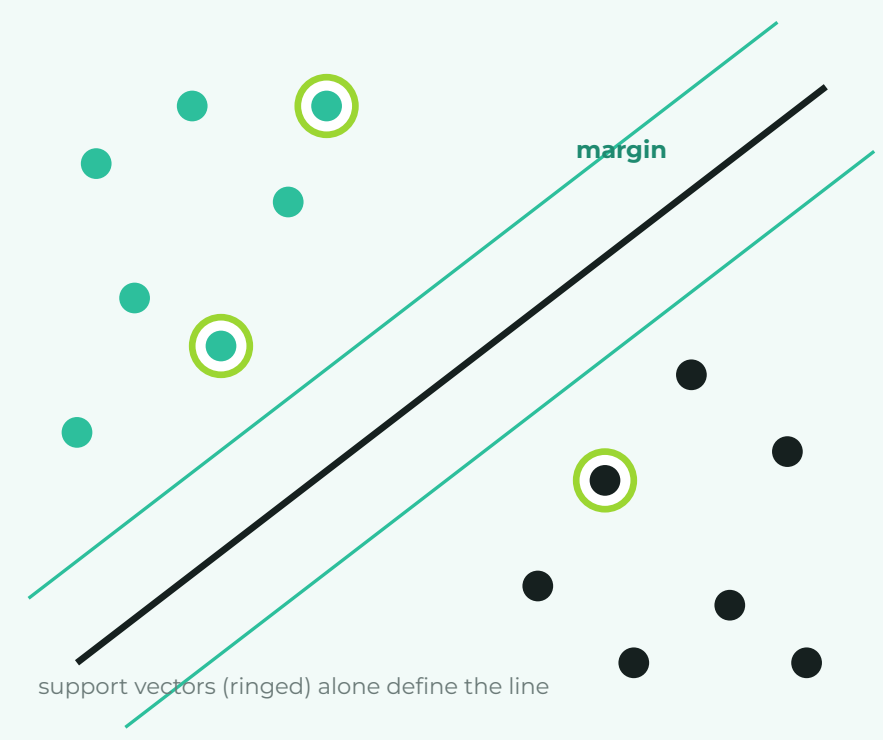
GOOD, BAD, 2026

- ✓ **No training:** a lazy learner that remembers everything.
- ✓ **Slow at test time:** compares with every example.
- ✓ **Explainable:** “these five posts look like yours”.
- ✓ **2026:** kNN over embeddings is exactly how vector databases and retrieval for LLMs work.

SOURCE Yang & Liu, SIGIR 1999; IIR Ch. 14.3; Farseev, ITMO SMA Module 2 (kNN algorithm and category thresholds). Points are illustrative.

CLASSIFIERS · SUPPORT VECTOR MACHINES

SVM: draw the widest possible street between classes.



THE MATHS, BRIEFLY

$g(d) = w^T d + b$

decision: the sign of g

$y_i(w^T d_i + b) \geq 1$

every post on its correct side

margin = $2 / \|w\|$

maximise it → minimise $\|w\|$

soft margin, C

tolerate some mistakes

kernels

curved boundaries in a higher space

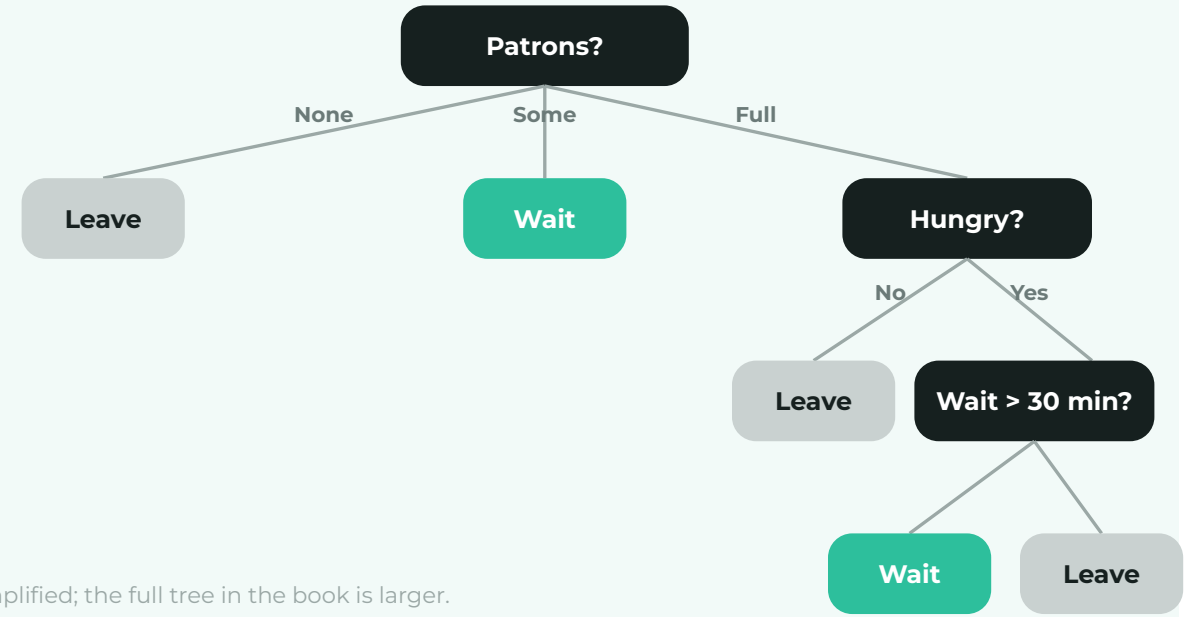
WHY IT SUITED TEXT

- ✓ **High-dimensional, sparse:** tens of thousands of word features barely hurt it.
- ✓ **Global optimum:** training is a convex problem.
- ✓ **Linear is enough:** most text categories are linearly separable (Joachims).
- ✓ **2026:** a linear SVM on tf.idf is still a baseline worth beating.

SOURCE Joachims, “Text categorization with Support Vector Machines”, ECML 1998; Cortes & Vapnik, Machine Learning 1995; IIR Ch. 15; Farseev, ITMO SMA Module 2.

A decision tree: twenty questions, learned from data.

WAIT FOR A TABLE? (AFTER RUSSELL & NORVIG)



Simplified; the full tree in the book is larger.

HOW IT IS LEARNED

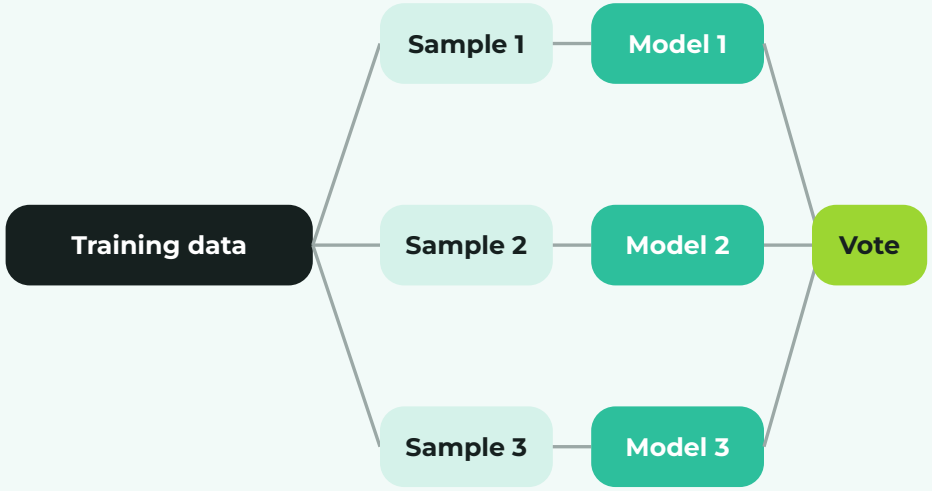
- ✓ **Greedy:** at each node, split on the feature with the highest information gain.
- ✓ **Stop:** when a node is pure, or at a depth limit.
- ✓ **Prune:** cut branches that only fit noise.

GOOD AND BAD

Readable rules you can show a client; handles mixed feature types; learns feature combinations (e.g. has video AND posted on Monday). But one tree is unstable and overfits easily — which is why we grow forests.

SOURCE Russell & Norvig, Artificial Intelligence: A Modern Approach (restaurant example); Quinlan, “Induction of decision trees”, Machine Learning 1986; Farseev, ITMO SMA Module 2.

Ensembles: many weak learners, one strong decision.



“Crowd-sourced” machine learning: combine by voting or averaging.

BAGGING

Train in parallel on bootstrap samples; vote or average. Cuts variance. → random forest.

BOOSTING

Train in sequence; each new learner fixes the last one's errors. AdaBoost; gradient-boosted trees (XGBoost).

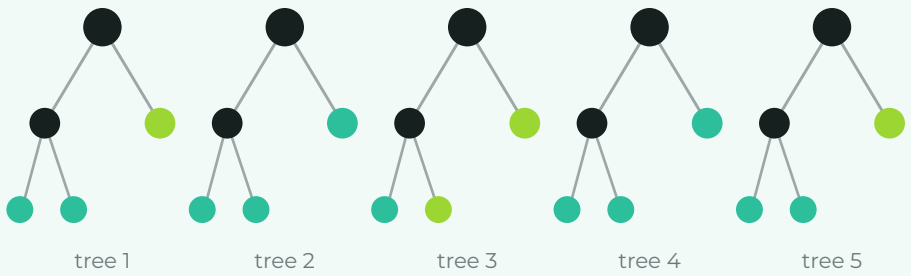
STACKING

A second-stage model learns how to combine the first-stage outputs — like late fusion in section 3.

SOURCE Breiman, “Bagging predictors”, 1996; Freund & Schapire, JCSS 1997 (AdaBoost); Chen & Guestrin, KDD 2016 (XGBoost); Wolpert 1992 (stacking); Farseev, ITMO SMA Module 2.

CLASSIFIERS · RANDOM FOREST

Random forest: bagged trees, each seeing a random slice.



Majority vote → teal

Two sources of randomness make the trees disagree in useful ways — and their errors cancel out.

01 Bootstrap rows

Each tree trains on a sample of posts drawn with replacement.

02 Random features

At each split, try only a random subset (default ≈ √ of all features).

03 Unpruned trees

Each tree overfits on purpose; averaging removes the noise.

Lab 1: your author and post features form exactly this kind of dense table — try a forest before anything fancier.

IN THIS COURSE

- ✓ **Strong default** for dense feature tables; little tuning.
- ✓ **Parallel:** trees train independently.
- ✓ **NUS-MSS:** RF beat SVM and Naive Bayes on every source.
- ✓ **Beaten by** boosted trees and neural nets at the top end.

SOURCE Breiman, “Random Forests”, Machine Learning 45(1), 2001; Farseev et al., ICMR 2015 (Table 3); Farseev, ITMO SMA Module 2.

TEXT CLASSIFICATION · 2026

Three ways to classify text today. Pick by cost and data, not by hype.

A · Features + classic model

tf.idf, lexicons, counts → random forest or logistic regression.

Labels needed	**Hundreds–thousands**
Cost per item	**Lowest**
Explain it	**Easy**

B · Embeddings / fine-tuned encoder

Sentence embeddings + a linear head, or fine-tune a small encoder.

Labels needed	**Hundreds+**
Cost per item	**Low**
Explain it	**Harder**

C · Prompt an LLM

Zero-shot or a few examples in the prompt; ask for one label.

Labels needed	**None**
Cost per item	**Highest; slow at scale**
Explain it	**Hard to audit**

THE LAB 1 LINK

The reference baseline is option C, a few-shot LLM. Options A and B can beat it with signals the prompt never sees: author history, timing, media, career data.

SOURCE Lab 1 brief (Lecture 1 appendix). Trade-offs are typical, not fixed: measure on your own data.



TEXT CLASSIFICATION · EMBEDDINGS

From counting words to embedding meaning.

“You shall know a word by the company it keeps.” J. R. Firth, 1957 — the distributional hypothesis behind every embedding.

2013

word2vec

Predict a word from its neighbours (CBOW), or the neighbours from a word (skip-gram). Google.

2014

GloVe

Factorise global word co-occurrence counts. Pennington, Socher, Manning — Stanford.

2018

BERT

Contextual vectors: “bank” differs in “river bank” and “bank loan”.

2019

Sentence-BERT

One vector per sentence. Most similar pair among 10,000 sentences: ~65 hours → ~5 seconds.

2023–25

MTEB → MMTEB

Benchmarks to choose a model: 500+ tasks, 250+ languages. No model wins everywhere.

Where to learn it: Stanford CS224N (Winter 2026), Lecture 2 “Word Vectors” · Jurafsky & Martin, SLP3 Ch. 5 “Embeddings” · MIT 6.S191 (2026), Lecture 2 “Deep Sequence Modeling”

SOURCE Stanford CS224N (W2026) L2 “Word Vectors”; Jurafsky & Martin SLP3 Ch. 5; MIT 6.S191 (2026) L2; Mikolov 2013; Pennington 2014; Devlin 2019; Reimers 2019; MMTEB 2025.

TEXT CLASSIFICATION · EMBEDDINGS IN PRACTICE

Embeddings for classification: a four-step recipe.

01

Embed

Run every post through one embedding model — the same model for train and test.

→

02

Normalise

Scale every vector to length 1: only its direction — the meaning — counts, not how long the post is.

→

03

Train a small classifier

Keep the embedding model frozen; train logistic regression to map each vector to a label.

→

04

Evaluate

Macro-F1 on held-out authors, always against a tf.idf baseline.

CHOOSING THE MODEL

- ✓ **Match the task:** check MTEB's classification scores, not retrieval.
- ✓ **Match the language:** Russian and code-switched posts need a multilingual model.
- ✓ **Size isn't everything:** MMTEB's best public multilingual model had 560M parameters.

FEW LABELS?

8 examples per class

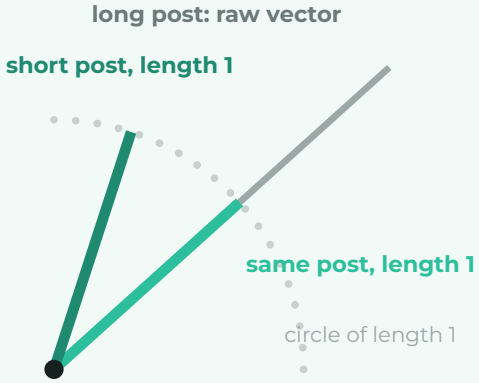
SetFit on customer-review sentiment was competitive with RoBERTa-Large fine-tuned on 3,000.

SOURCE Stanford CS224N (W2026) L2; Reimers & Gurevych, EMNLP 2019; Tunstall et al., SetFit, 2022 (+ Hugging Face blog); Enevoldsen et al., MMTEB, ICLR 2025.

TEXT CLASSIFICATION · STEPS 2 AND 3, EXPLAINED

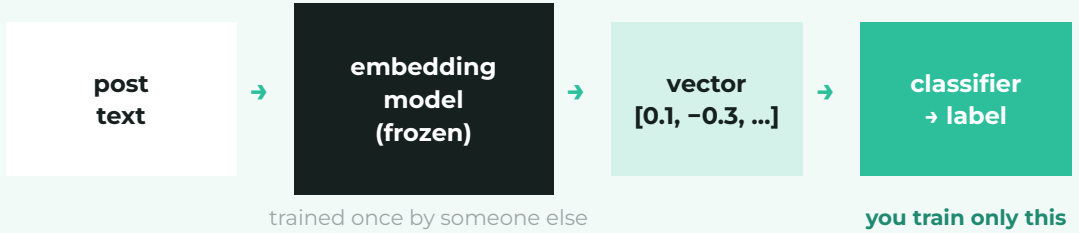
What “normalise” and “train a classifier on top” mean.

STEP 2 · NORMALISE: KEEP THE DIRECTION, DROP THE LENGTH



Divide each vector by its length. A 300-word post and a 20-word post on the same topic now point the same way and compare fairly. Then the angle between two vectors — cosine similarity — is just a dot product: fast to compute for millions of posts.

STEP 3 · A SMALL CLASSIFIER ON TOP OF FROZEN EMBEDDINGS



- ✓ **Why logistic regression:** one weight per vector dimension; trains in seconds; outputs a probability per class.
- ✓ **Why frozen:** the big model already knows language; you only teach the last, small step.
- ✓ **Very few labels?** SetFit first nudges the embedding model with pairs of same-class and different-class posts, then trains the classifier.

SOURCE Stanford CS224N (W2026) L2 (cosine similarity of word vectors); Jurafsky & Martin SLP3 Ch. 4 (logistic regression) and Ch. 5; Tunstall et al., SetFit, 2022.

TEXT CLASSIFICATION · EVALUATION

Accuracy can lie. Macro-F1 makes every class count.

- ✓ **Precision.** Of what we labelled positive, how much was right?
- ✓ **Recall.** Of what was really positive, how much did we find?
- ✓ **F1.** Harmonic mean of the two; punishes being good at only one.
- ✓ **Micro-average.** Pool every prediction. Big classes dominate.
- ✓ **Macro-average.** F1 per class, then a plain mean. Every class counts equally.

90% ADULTS, 10% TEENS

A model that always says “adult”:

90%

accuracy

47%

macro-F1 — it never finds a single teenager. Lab 1 is scored on macro-F1.

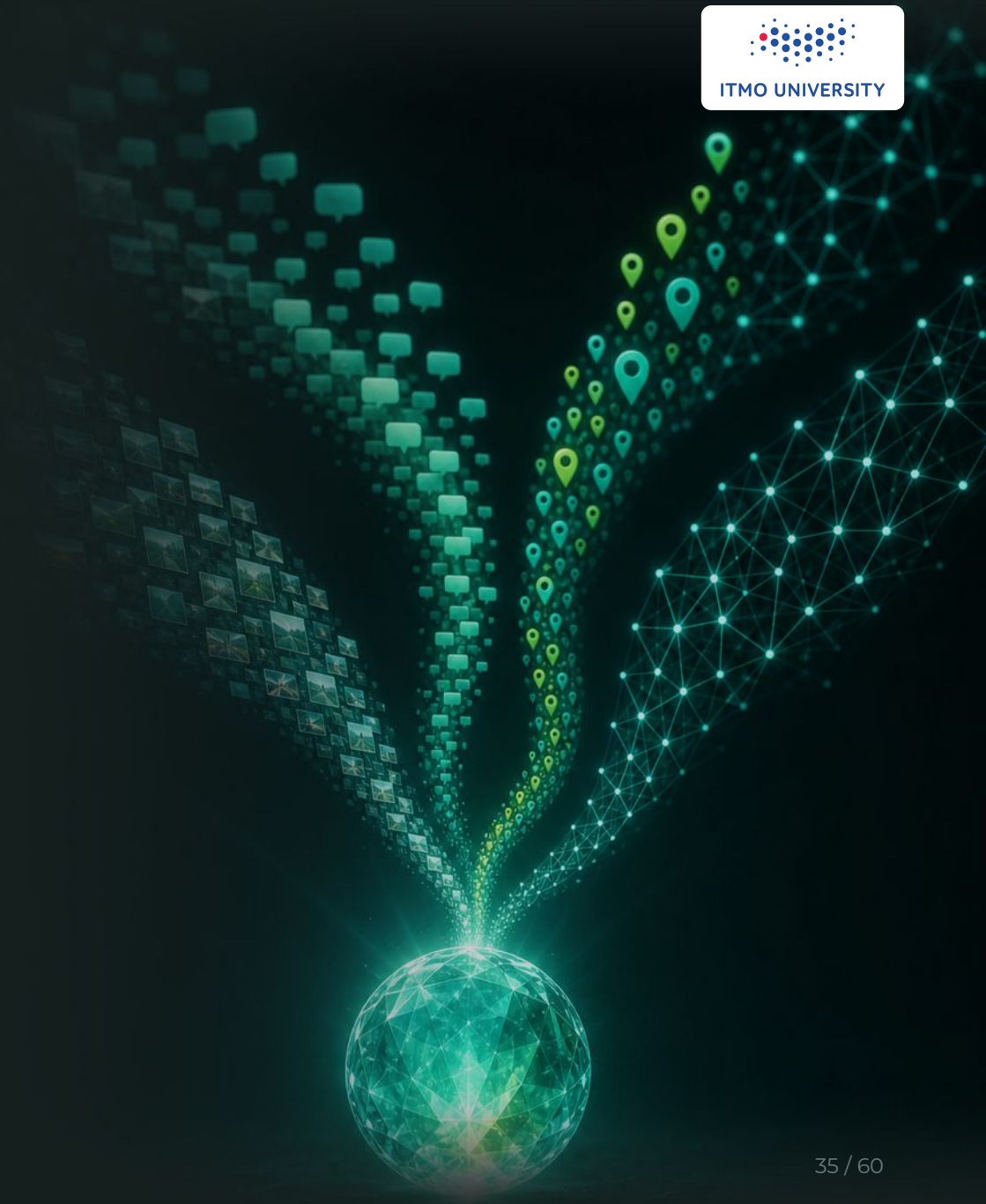
SOURCE Farseev, ITMO SMA Module 2 (P/R/F, micro vs macro); worked example: adult F1 = 0.947, teen F1 = 0, mean = 0.47.

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03

SECTION 3

Multi-source fusion



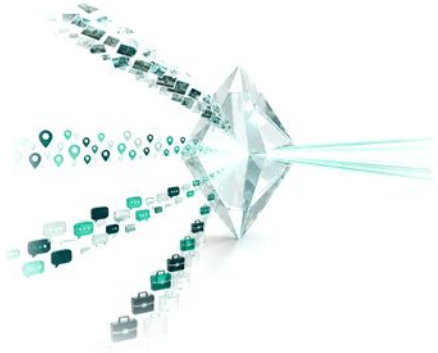
DATAREPORTAL · DIGITAL 2026

The average internet user is on 6.75 social platforms every month.

Each one sees a different side of the same person. None of them sees all of it. Combining them well is harder — and more interesting — than it sounds.

FUSION · ONE USER, MANY VIEWS

Each platform sees one side of you.



PROFESSIONAL NETWORK

The formal self: career, education, skills.

PUBLIC MICROBLOG

The casual, opinionated self.

VISUAL PLATFORM

Interests and lifestyle through images.

MAPS & LOCATION

Routines: home, work, gym, Saturday café.

HOW SOURCES RELATE

Complementary

Two cameras with different fields of view: each adds what the other misses.

Redundant / consistent

Age and gender are the same everywhere, so sources can check each other.

Cooperative

Only meaningful together, like the audio and video of one clip.

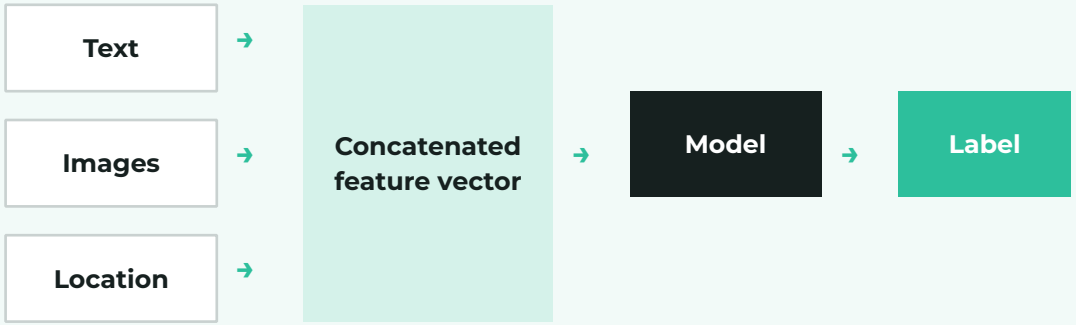
SOURCE Castanedo, “A Review of Data Fusion Techniques”, The Scientific World Journal 2013; Farseev, ITMO SMA Lecture 8.

FUSION · STRATEGIES

Early fusion joins features. Late fusion joins opinions.

EARLY FUSION

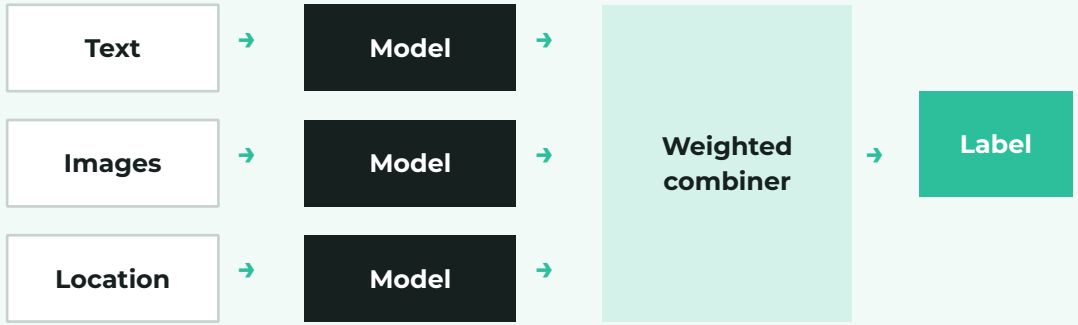
One long vector, one model



Risk: huge vectors; big sources drown small ones; a missing source leaves a hole.

LATE FUSION

One expert per source, then combine



Gain: each expert weighted by confidence and data volume; missing sources just drop out.

INTERMEDIATE FUSION Encode each source, fuse inside the model. That is how 2026 multimodal LLMs read text and images together.

SOURCE Farseev, ITMO SMA Lecture 8 (early vs late fusion); Farseev et al., ICMR 2015.

FUSION · THE MISSING-SOURCE PROBLEM

Most users are not on every platform. Fill the holes carefully.

	Career	Posts	Photos	Places
User 1	✓	✓	?	✓
User 2	✓	?	✓	?
User 3	?	✓	✓	✓
User 4	✓	✓	?	?
User 5	?	✓	✓	✓

- **Nearest-neighbour imputation.** Copy the value from the most similar user. Simple; noisy.
- **Matrix factorisation.** Each user has a small hidden vector; learn it from the cells you see, predict the rest.
- **Shared latent space.** One hidden vector across sources, so a career profile helps fill the photo column.
- **Coming back in Lecture 5.** The same idea powers recommender systems.

SOURCE Song, Nie, Zhang, Akbari, Chua, “Multiple social network learning and its application in volunteerism tendency prediction”, SIGIR 2015.



FUSION · CASE: TWEETFIT (AAAI 2017)

Can social posts and a fitness tracker reveal your BMI?

THE QUESTION

Infer a wellness profile

Two attributes, from four sources at once:

BMI category — 8 WHO classes, from severe thinness to obese III.

BMI trend — is weight going up or down?

SOURCE	MODALITY	POSTS	USERS	BMI LABELS
Twitter	text	1,676,310	5,375	1,052
Endomondo	sensors	140,926	4,205	870
Instagram	images	48,137	2,062	372
Foursquare	places	19,743	609	116

Imbalanced: 38% “normal”, only 3% “moderate thinness”; 56% trending down.

NUS-SENSE DATASET · MAY–AUG 2015

Ground truth came from public Endomondo profiles, where athletes log height and weight: BMI is weight over height squared, and the trend is the change between the first and last log. Accounts were linked because users shared their workouts to Twitter.

SOURCE Farseev & Chua, “TweetFit: Fusing Multiple Social Media and Sensor Data for Wellness Profile Learning”, AAAI 2017, pp. 95–101 (Table 1).

FUSION · TWEETFIT: HOW IT WORKS

One feature set per source, one multi-task model for all users.

STEP 1 · FEATURES PER SOURCE

Text	50 LDA topics · writing style (mistakes, slang, sentiment) · food & health lexicons
Places	764 venue categories → 86 PCs · mobility: posting by time of day, areas of interest
Images	1,000 ImageNet concepts → 150 PCs
Sensors	distance, speed, duration · 96 workout types · FFT of heart rate, speed, altitude, cadence, oxygen: 495 → 54 PCs

STEP 2 · USERS WITH THE SAME SOURCES FORM ONE TASK



Toy example from the paper. Missing sources never need filling in — each combination is its own task.

W

one weight column per task; rows = features

ℓ2,1: a source's features are kept or dropped together, for every task

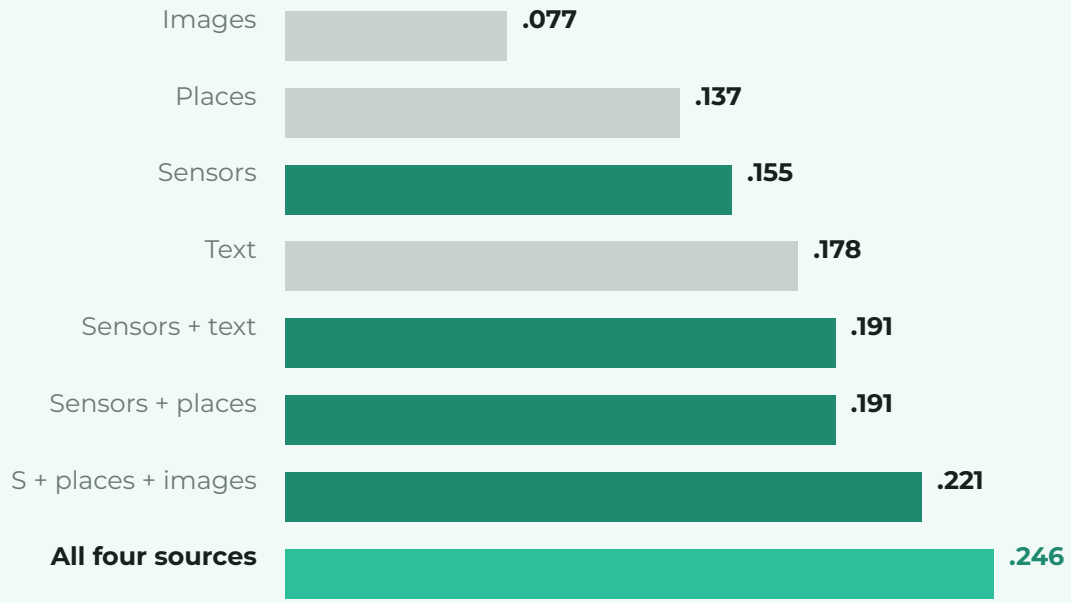
min $\Psi(X, W, Y) + \lambda \|W\|_{2,1}$ logistic loss over all tasks + group sparsity; solved with Nesterov's accelerated gradient

SOURCE Farseev & Chua, AAAI 2017 (Sec. “Individual Wellness Profiling”, Eq. 1, Fig. 3); multi-task learning: Caruana 1997; ℓ2,1 regularisation: Liu, Ji & Ye 2009.

FUSION · TWEETFIT: RESULTS

All four sources won — and sensors were in every top mix.

BMI CATEGORY · MACRO-F1 (TABLE 2) · DARK TEAL = INCLUDES SENSORS



AGAINST BASELINES (TABLE 3) · MACRO-F1

Method	BMI class	BMI trend
MSE ensemble (2015)	.142	.644
Random forest	.169	.480
iMSF	.172	.649
MTFL	.184	.710
TweetFit	**.211**	**.718**

+38% over the best single source (.246 vs .178). The authors add that accuracy is not yet good enough for real-world use — and health is a “special category” under GDPR (section 5).

SOURCE Farseev & Chua, AAAI 2017, Tables 2–3 (80/20 user split; Table 2 varies the sources, Table 3 compares methods). +38% computed from Table 2.

04

SECTION 4

Profiling in practice



J. W. PENNEBAKER, THE SECRET LIFE OF PRONOUNS, 2011

“The smallest … most forgettable words serve as windows into our thoughts, emotions, and behaviors.”

People reveal themselves less in what they say about themselves than in how they say everything else. That insight became LIWC — and it is why profiling from text works.

PROFILING · CASE: NUS-MSS (ICMR 2015)

One user, three networks, three cities.

CITY	USERS	TWEETS	CHECK-INS	IMAGES
Singapore	7,023	11,732,489	366,268	263,530
London	5,503	2,973,162	127,276	65,088
New York	7,957	5,263,630	304,493	230,752
Total	**20,483**	**19,969,281**	**798,037**	**559,370**

Sources: Twitter (text), Foursquare (location), Instagram (images). Ground truth for gender and age from public profile pages. Crawled Jul–Dec 2014.

✓ **Privacy by release:** only extracted features were published, never raw posts.

THE IDENTITY TRICK

Cross-posting linked the accounts.

In 2014 people shared Instagram photos and Swarm check-ins to Twitter. Each shared link tied two accounts to one person.

A historical case: not provided in this course. APIs closed, cross-posting gone — Lab 1 uses our LinkedIn corpus.

SOURCE Farseev, Nie, Akbari, Chua, ICMR 2015 (Table 1); Farseev & Chua, NUS-MSS dataset (2015).

PROFILING · FEATURES BY SOURCE

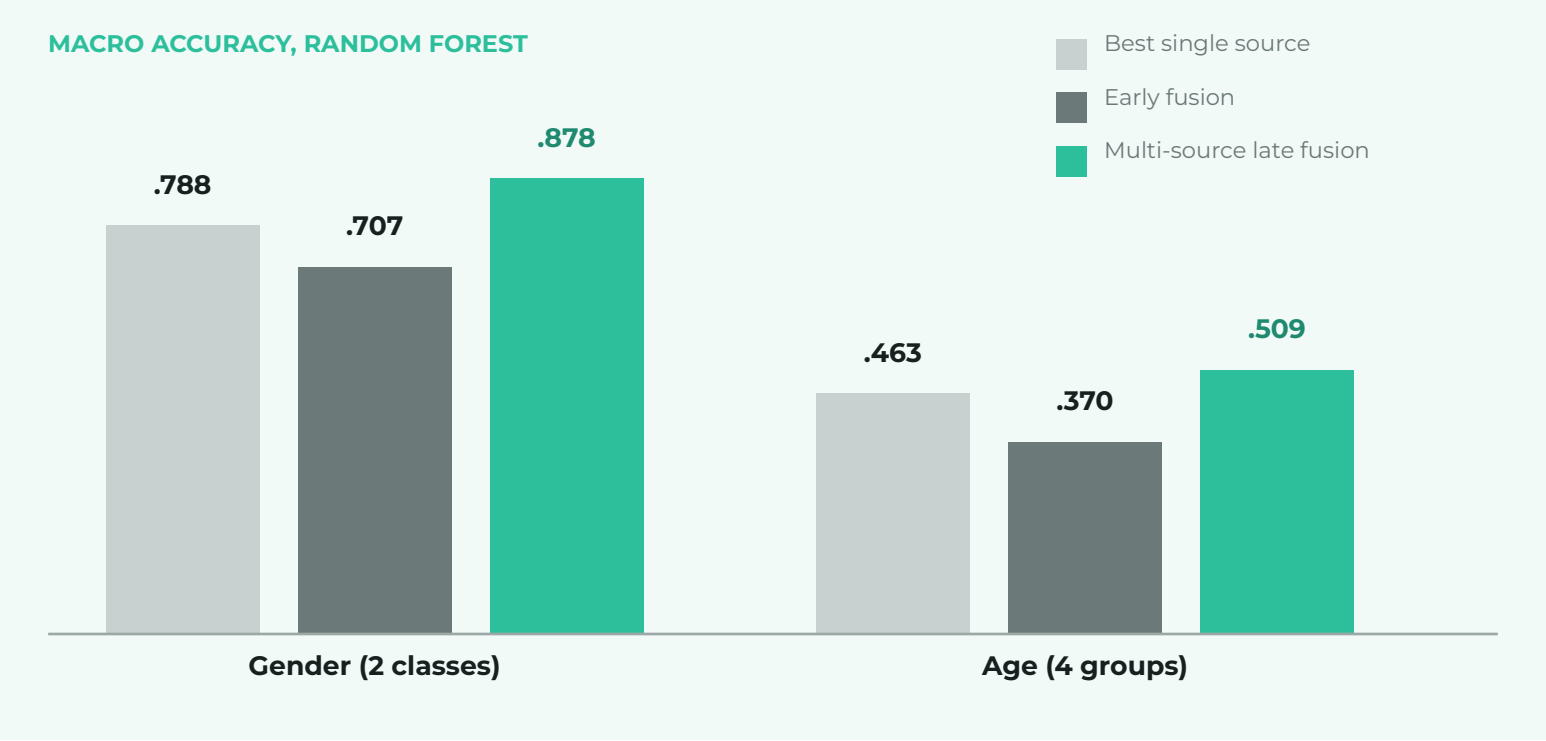
What each source contributed in 2015 — and the 2026 equivalent.

SOURCE	2015: NUS-MSS FEATURES	2026 EQUIVALENT
Text	LIWC (71 psycho-linguistic categories: “I” vs “we”, emotion words). 50 LDA topics. Writing behaviour: hashtags, slang, “noooo”, emoticons, misspellings.	**LIWC-22 (100+ dimensions, 12k+ entries). Sentence embeddings for topic and style. LLM-extracted attributes.**
Location	Venue-category vector over 592 Foursquare categories: 2 restaurants + 1 airport → ⅔ restaurant, ⅓ airport. 6 mobility topics.	**Check-ins faded. Map reviews, geo-tagged video, on-device location — and what a VLM infers from photos.**
Images	1,000 ImageNet concepts from SIFT / colour-histogram detectors (≈64% accuracy on ILSVRC 2012).	**CLIP-style image embeddings; vision-language models (Gemini, GPT) that describe a photo in words.**

SOURCE Farseev et al., ICMR 2015 (Sec. 4); Boyd, Ashokkumar, Seraj, Pennebaker, LIWC-22 manual (2022).

PROFILING · RESULTS

Late fusion beats every single source. Early fusion can hurt.



TWO LESSONS

+9 pts

late fusion over the best single source for gender (.878 vs .788).

More data helps only if you fuse it well. Early fusion (.707) lost to text topics alone.

Evaluated on 222 Singapore users active on all three networks.

SOURCE Farseev, Nie, Akbari, Chua, ICMR 2015, Table 3. Best single source: LDA topics (gender), writing behaviour (age).

PROFILING · WHAT THE SOURCES REVEAL

The best source depends on the question.

Gender	Images were strongest.
Age	Location was strongest.
Both	Text helped every attribute.

SIX MOBILITY TOPICS FROM SINGAPORE CHECK-INS

Food lovers
Restaurants, malls

Business travellers
Cafés, airport, hotels

Party goers
Nightclubs, food courts — mostly under 20

Family people
Home, office, neighbourhood

Students
University, gym, fitness

Teenagers
Stations, malls, high schools

READ IT LIKE AN ANALYST

The old slide said women photograph flowers and men photograph cars. These are averages over one 2014 sample, not facts about people. A profile is a probability — never an identity.

SOURCE Farseev et al., ICMR 2015 (Sec. 5–6, mobility topics via LDA over venue categories).

PROFILING · MULTIMODAL SOURCES

Every platform is already multimodal.



TIKTOK

Video frames, voice, music, on-screen text, captions

Age cues, humour, trends; places from the scenery.

INSTAGRAM

Photos, captions, hashtags, geotags, tagged friends

Lifestyle, interests, places, the social circle.

YOUTUBE

Thumbnails, titles, transcripts, comments, watch time

Expertise, audience, sentiment of the crowd.

X / THREADS

Short text, replies, reposts, the follow graph

Opinions, politics, communities, influence.

MAPS & REVIEWS

Review text, photos, ratings, visit history

Mobility, routines, spending level.

TELEGRAM CHANNELS

Long posts, forwards, reactions, subscriber growth

Topic authority; networks of channels.

XIAOHONGSHU (REDNOTE)

Notes mixing photos, text and product tags

Shopping intent, taste, life stage.

LINKEDIN

Profile, career timeline, posts, documents, reactions

Next: one member, taken apart — the same shape as Lab 1's data.

Section 3 in one line: one account is already several sources — text, images, sound, place, network — before you ever join two platforms.

SOURCE Platform features as of 2026; modality-to-attribute mapping follows Farseev et al., ICMR 2015 and Lecture 1 (platform landscape).

PROFILING · CASE: ONE LINKEDIN MEMBER

One professional profile holds six sources.



A7

AUTHOR_0412

Head of Growth · B2B SaaS · writes about AI in marketing

Singapore · 2,310 followers

1	**About**	“I help SaaS teams grow with data and AI…”
2	**Experience**	ORG_117 Head of Growth 2022– · ORG_052 Analyst → Senior 2016–22
3	**Education**	UNI_008 · BSc Economics · 2012–2016
4	**Skills**	Growth marketing · SQL · A/B testing · LLMs
5	**Activity**	212 posts · 38% with images, video or PDF carousels
6	**Network**	Reactions: Like, Celebrate, Insightful… · who reacts, who follows

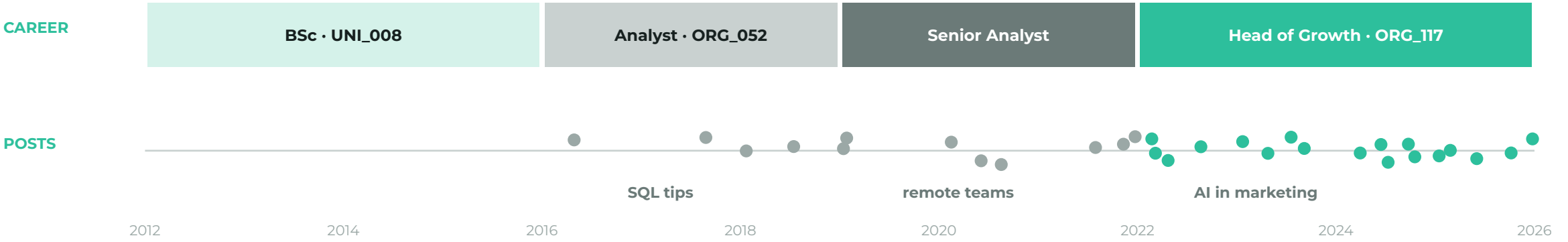
SOURCE	FEATURES	PREDICTS
1 **About text**	embeddings, lexicons (sec. 2)	**occupation, interests, tone**
2 **Career timeline**	structured + temporal features	**seniority, tenure, job changes**
3 **Education**	graduation year	**age band — the NUS-MSS trick**
4 **Skills**	named entities, lexicons	**expertise, industry**
5 **Posts + media**	text features, VLM, OCR on slides	**topics, style, personality**
6 **Engagement & network**	reaction mix, followers, homophily	**influence, authenticity**

SOURCE Illustrative, pseudonymised profile in the format of the Lab 1 corpus (codes like ORG_117 replace names). Reaction types as in the Lab 1 brief.

PROFILING · CASE: THE TIMELINE

A career timeline is a profile that changes over time.





01

Age band

Graduated 2016 → born ≈1994 → 30–34. NUS-MSS inferred age from education too (±2.3 yrs).

02

Life events

Promotion 2019, new company 2022: “I'm happy to share…” posts. Moments marketers pay for.

03

Topic drift

SQL → remote teams → AI. Weight recent posts higher: the evolving-text features of section 2.

04

No peeking

Predict each post from strictly earlier data only — the Lab 1 rule that stops leakage.

SOURCE Illustrative timeline, pseudonymised as in Lab 1; Farseev et al., ICMR 2015 (education-based age estimation); Chen, Li, Chua 2014 (temporal term weighting).

PROFILING · CASE: FUSING THE SOURCES

Which source answers which question? Fuse by confidence.

	About text	Career	Post text	Post media	Engagement	Network
Age band	○	●	○	○	–	○
Occupation & seniority	●	●	○	–	–	○
Interests & topics	○	–	●	●	○	○
Personality	–	–	●	○	○	–
Influence	–	○	–	–	●	●
Authenticity: human, bot, ghostwritten?	○	○	●	○	●	●

● strong ○ some signal

LATE FUSION, APPLIED

- ✓ **One expert per column,** as in NUS-MSS and TweetFit.
- ✓ **Weight by confidence:** career is the best witness for seniority.
- ✓ **Weight by volume:** 200 posts count for more than 5.
- ✓ **Missing source?** No media — that expert drops out.
- ✓ **Check consistency:** a 2016 graduate claiming “20 years in the industry” is a flag.

SOURCE Illustrative strengths, not measured values. Method: Farseev et al., ICMR 2015 (weighted late fusion); Song et al., SIGIR 2015 (source confidence & consistency).

PROFILING · FROM THE CASE TO LAB 1

Lab 1 is this LinkedIn case — 5,439 times over.

151,281

POSTS

Text, date, engagement; images, video and slide documents.

5,439

AUTHORS

Real professionals, a median of 24 posts each.

55,506

POSTS WITH MEDIA

42,700 with images, 10,787 with video, 2,019 slide decks.

THE SIX SOURCES, IN THE LAB 1 FILES

- ✓ **Posts:** text, date, seven reaction counters, mentions, company tags, reshares.
- ✓ **Media:** type and count on every post; extracted text of slide decks.*
- ✓ **Author profile:** career history and skills.
- ✓ **Followers:** for 98.7% of authors, with the moment each count was read.
- ✓ **Stable codes:** one company keeps one code in posts and careers.
- ✓ **Pseudonymised:** no names, handles or profile photos.

LAB 1 IN ONE LINE

Profile the author from their earlier posts and career — then judge this post.

- ✓ **Label:** below, typical or above the author's usual.
- ✓ **Split by author:** 3,584 train, 896 held out.
- ✓ **Released today;** due Sun 25 Oct, 23:59.

SOURCE Lab 1 brief, Lecture 1 (23 Sep 2026), slides 33–37; EDRA corpus. * Which media files ship depends on the anonymisation review.

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PROFILING · PERSONALITY

Can a model know you better than your friends?



O Openness	**C** Conscientiousness	**E** Extraversion	**A** Agreeableness	**N** Neuroticism

2013 · KOSINSKI ET AL., PNAS

AUC 0.93

Facebook likes of 58,466 volunteers separate men from women; Democrats vs Republicans AUC 0.85.

2015 · YOUYOU ET AL., PNAS

300 likes

to judge personality better than a spouse. 10 beat a colleague, 70 a friend, 150 a family member.

2024 · PETERS & MATZ, PNAS NEXUS

r = 0.29

GPT-4, zero-shot, from Facebook status updates of 1,000 users — on par with models trained for the job.

SOURCE Kosinski, Stillwell, Graepel, PNAS 2013; Youyou, Kosinski, Stillwell, PNAS 2015; Peters & Matz, PNAS Nexus 3(6) 2024.

PROFILING · LLMS AS PROFILERS

One model now profiles from text and photos.



TEXT · ETH ZURICH, ICLR 2024

84.6%

top-1 accuracy, 8 attributes, 520 Reddit profiles.

- ✓ **Sex ≈98%.** Income hardest ≈63%.
- ✓ **Anonymised text:** location still inferred ≈55% of the time.

PHOTOS · "DOXING VIA THE LENS", 2025

60%

of o3's location guesses for 50 street photos of people fell within 1 mile.

- ✓ **84%** within 5 miles.
- ✓ **Give-aways:** street layout and front gardens.

WHAT CHANGED

The cost of profiling fell to nearly zero.

2015: a dataset, features and a trained model per attribute.
2026: one prompt, any attribute, any user with public posts.

SOURCE Staab, Vero, Balunović, Vechev, "Beyond Memorization", ICLR 2024; "Doxing via the Lens", arXiv 2504.19373 (Apr 2025).

PROFILING · MOBILITY

Where you go is highly predictable.



1885

Laws of migration

Ravenstein: most people move short distances; long-distance movers pick big cities.

2006

Where's George?

Tracked banknotes: travel distance follows a power law, exponent ≈ 1.59.

2008

100,000 phones

González et al.: people return again and again to a few places.

2010

93% predictable

Song et al., Science: the potential predictability of where you'll be.

2026

The data moves away

Google moved Maps Timeline onto phones and removed the web version.

DEMOGRAPHIC MOBILITY, SINGAPORE (NUS-MSS)

Teenagers check in near homes and schools; students and young professionals cluster in the city centre; adults take weekend trips to nearby cities.

SOURCE Ravenstein, JRSS 1885; Brockmann, Hufnagel, Geisel, Nature 2006; González, Hidalgo, Barabási, Nature 2008; Song et al., Science 2010; Google Maps Timeline (2024–25); Farseev et al., ICMR 2015.

ITMO UNIVERSITY

05

SECTION 5

Where profiling goes too far



META · 19 JANUARY 2022

Meta removed ad targeting based on health, race, politics, religion and sexual orientation.

The categories were inferences about sensitive traits — and they had been misused. The question is no longer can we profile, but should we, and who decides.

LIMITS · PITFALLS

Five ways profiling goes wrong — even when the model is accurate.

01

Correlation ≠ identity

A 70%-confident guess is wrong 3 times in 10 — and always wrong for those people.

02

Stereotypes in, out

Biased training data teaches the model to reward the stereotype.

03

Proxy labels

Ground truth is scarce; NUS-MSS estimated many ages from education history.

04

Feedback loops

The profile picks what you see; what you see shapes the next profile.

05

Sensitive inference

Health, sexuality, religion, politics — inferred from harmless data. The inference itself can harm.

Ask of every profile: who is it wrong for, how often, and what happens to them when it is?

SOURCE Farseev et al., ICMR 2015 (age ground truth); Kosinski et al., PNAS 2013 (sensitive-trait inference).

LIMITS · THE RULES IN 2026

The law draws lines around sensitive inference.



GDPR · ART. 9

Special categories

Health, ethnicity, religion, politics, sexual orientation need a strong legal basis.

DSA · ART. 26(3), 28(2)

No sensitive-data ad profiling

Since Feb 2024: no ads profiled on special categories; none profiled on known minors.

AI ACT · ART. 5(1)(G)

Biometric categorisation ban

Since Feb 2025: no inferring race, politics, religion or orientation from biometrics.

DMA · APR 2025

€200M fine for Meta

Pay-or-consent ruled non-compliant; EU users now get a less-personalised-ads option.

FOR YOUR PROJECT

- ✓ **Say what you infer** and why.
- ✓ **Show it** to the user.
- ✓ **Report errors** by group, not only overall.
- ✓ **Beyond the EU:** Singapore PDPA, Russia 152-FZ.

SOURCE GDPR Art. 9; DSA Art. 26(3), 28(2); EU AI Act Art. 5(1)(g); European Commission DMA decision (23 Apr 2025).

COURSE · FROM LECTURE TO LAB

Everything from today is a tool for Lab 1.

LAB 1 · RELEASED TODAY · DUE SUN 25 OCT, 23:59

LinkedIn: did this post beat its author's usual?

- **Text features:** tf.idf, lexicons, embeddings.
- **Feature selection:** the JSON is huge.
- **Macro-F1:** no majority-class shortcut.
- **A second source:** fuse the author profile — career, skills, followers — with the post.

LAB 2 · DUE SUN 6 DEC

Individual. A harder task fusing text, images and behaviour — e.g. demographics, personality or virality.

PROJECT IDEAS

- **An age-assurance classifier that explains itself.**
- **A privacy auditor: what can an LLM infer from my posts?**
- **Fair profiling: error rates reported by group.**
- **A bot-vs-human feed classifier.**

SOURCE Lab 1 brief (Lecture 1 appendix, slides 33–37); course schedule (Lecture 1, slide 30).

COURSE · GO DEEPER

Free reading from Stanford, MIT and the papers behind today.

01 TEXT CLASSIFICATION

Jurafsky & Martin — Speech and Language Processing (3rd ed. draft, Aug 2026): Ch. 4 Logistic Regression & Text Classification; App. B Naive Bayes

web.stanford.edu/~jurafsky/slp3

02 EMBEDDINGS

Stanford CS224N (Winter 2026) — Lecture 2: Word Vectors (word2vec, GloVe, evaluation)

web.stanford.edu/class/cs224n

03 DEEP LEARNING

MIT 6.S191 (2026) — Lecture 2: Deep Sequence Modeling (embeddings, attention, language models)

introtodeeplearning.com

04 MULTI-SOURCE PROFILING

Farseev, Nie, Akbari, Chua — Harvesting Multiple Sources for User Profile Learning, ACM ICMR 2015

dl.acm.org — ICMR 2015, pp. 235–242

05 LLM PRIVACY

Staab et al. — Beyond Memorization: Violating Privacy via Inference with LLMs, ICLR 2024

arxiv.org/abs/2310.07298

06 CHOOSING AN EMBEDDING MODEL

MTEB leaderboard — Massive Text Embedding Benchmark (MTEB, EACL 2023; MMTEB, ICLR 2025)

huggingface.co/spaces/mteb/leaderboard

OTHER DETAILS · NEXT LESSON

Next week: Lecture 3 — networks and influence.



- ✓ **Lab 1:** released after class on the course Telegram channel. Due Sun 25 Oct, 23:59.
- ✓ **Office hours:** Prof. Aleks Farseev, 12 PM on Zoom, via Daniel Onishchenko or Jack Zhao.
- ✓ **Labs & project:** questions go to the tutors, Daniel Onishchenko and Jack Zhao.
- ✓ **Slides & reading:** posted on Telegram after class.

Next week: from single users to the connections between them — and why followers are not influence.

BEFORE LECTURE 3

Profile ten authors by hand.

Read ten posts from the Lab 1 data and guess about each author — before any model. Then mark which guesses came from the words, and which from your assumptions.